PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Equipment list and mechanical datasheets

PVC-V1-ME-EQL-001

Equipment list of the R-101 area (20 items from PFD-100) with design and operating conditions, the R-101 wall-thickness calculation, E-101 rating, R-101 nozzles and general arrangement, 37 checks against the model and the other documents (33 ✓ / 3 ✗), and datasheets of R-101, E-101, P-101A/B, D-102 and D-103 in Appendix A.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
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Originator	Apinan Soottitantawat

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The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Scope and basis

This document lists every item of equipment on PFD-100 for the R-101 area of the PVC Suspension Reactor Ver 1 OTS with its design and operating conditions, and gives the mechanical datasheets of R-101 (vessel, jacket, agitator), E-101, P-101A/B, D-102 and D-103 in Appendix A. It also records the R-101 wall-thickness calculation (ASME VIII Div. 1, indicative), the E-101 rating from the model's own relation, the R-101 nozzle list and general arrangement, and the checks of this document against the simulator model and the other documents of the set.

• Operating values: the design state 2.0 h into the reaction (snapDesign) or the reference batch (start of reaction at 57 °C, 60 t water + 51 t VCM, end point at 5.27 h), as stated.

• Source letters in the equipment list: **M** simulator model (constant, run or state), **D** another OTS document (P&ID PVC-PID-001/002, PFD-100, Design or Safety-layers tab), **C** computed here from M/D values, **A** assumed (typical practice).

• Design pressures follow the rule used on the Design tab: design ≥ max(1.1 × highest operating pressure, highest operating pressure + 1.7 bar) and ≥ the relief-valve set; design temperatures ≥ highest operating temperature + 15–30 °C.

• Standards are quoted as typical practice: ASME VIII Div. 1 (vessels), TEMA (exchangers), API 610 (pumps).

2 Design conditions of R-101

Quantity	Operating (source)	Design	How the design value is chosen
Internal pressure	8.62 bar g at 2.0 h; 8.88 bar g highest (cold batch, Transition tab)	15.2 bar g	Above the PSV-100 set (15 bar g) and PSHH-101 (12.5 bar g); the hottest grade (70 °C, 11.9 bar g) needs ≥ 13.6 bar g
External pressure	0.03 bar a when evacuated; jacket up to 5.2 bar g	FV + jacket	Evacuated before every charge: full vacuum, plus the jacket pressure on the shell (6.21 bar)
Temperature	57.0 °C; 58.21 °C highest (cold batch, Transition tab)	93 °C	≥ highest + 25 °C and ≥ 80.3 °C, the temperature at which the contents reach the PSV set
Jacket	loop ≈ 7.1 bar g, 38.2–40.9 °C; 86.6 °C in steam heat-up	5.2 bar g / 93 °C	P&ID-001 note 3; see engineering queries

3 Equipment list

Tag	Name	Type	Service	Design P bar g	Design T °C	Oper. P bar g	Oper. T °C	Volume m³	Size	Material	Motor kW (absorbed)	Notes
TK-100	VCM Storage	Horizontal drum, 2:1 heads	VCM day storage for the charge	8.5 C	65 A	2.97 M	25.0 M	77.8 D	ID 3.0 m × 10 m overall	CS impact-tested (SA-516 Gr 70N), MDMT −15 °C A	—	Holds one charge: 51 t VCM = 56.5 m³ at 25 °C (72.7 % full). Design P = max(1.1 P, P + 1.7 bar) on the VCM vapour pressure at 50 °C (6.8 bar g). MDMT covers auto-refrigeration of VCM to its boiling point −14 °C on depressuring.
P-103	VCM Charge Pump	Centrifugal, sealless (canned motor)	VCM from TK-100 to R-101 (FIC-102)	18.0 C	65 A	11.99 M	25.0 M	—	199.6 m³/h (180 t/h) × 9 bar	CS impact-tested casing, 316 SS impeller A	75 C (55 M)	Discharge 12 bar g (model K.Pvcm 13 bar a), suction 3 bar g. Estimated shut-off 13.8 bar g at 25 °C suction, 17.6 bar g with the storage at 50 °C (shut-off = suction + 1.2 × differential, typical API 610 curve).
TK-107	PVA Pot	Vertical pot, agitated	Suspending agent (PVA) solution into the water charge (HV-107)	3.5 A	80 A	0.00 A	30.0 A	2.0 A	≈ 2 m³	304L SS A	2 A	Dose per batch from the recipe; size and pressure assumed (not in the model).
D-104	Initiator Pot (chilled)	Vertical pot, jacketed for chilled brine	Fast + slow peroxide initiators, dosed by P-104 (XV-103)	3.5 A	50 / −20 A	0.30 A	-10.0 A	1.0 A	≈ 1 m³	316L SS A	—	Kept well below the self-accelerating decomposition temperature of the fast initiator (typical −15 to −5 °C). N_2 blanket; size assumed.

Tag	Name	Type	Service	Design P bar g	Design T °C	Oper. P bar g	Oper. T °C	Volume m³	Size	Material	Motor kW (absorbed)	Notes
P-104	Initiator Dosing Pump	Diaphragm metering pump	Initiator charge into R-101 in 60 s	20.0 A	50	10.62 C	-10.0	—	≈ 6 m³/h for 60 s	316L SS A	2 A	Must deliver against the running reactor pressure (8.6 bar g at 2.0 h, model). Dose time 60 s from the model; capacity assumed.
D-105	Short-stop Pot (N2)	Vertical pot, N_2-pressurised	Short-stop charge, injected by XV-104 on a trip	25.0 A	65 A	20.00 A	30.0	1.0 A	≈ 1 m³	316L SS A	—	N_2 pressure must exceed the PSV-100 set (15 bar g) so the short-stop goes in at any reactor pressure; PT-105 with a low alarm is recommended (Safety layers). Pressure and size assumed.
R-101	PVC Reactor 140 m3	Vertical jacketed stirred batch reactor, 2:1 heads, 4 cooled baffles	Suspension polymerisation of VCM (1 of 4: R-101…R-104)	15.2 / FV M	93 M	8.62 M	57.0 M	141.5 M	ID 4500 × 7400 T/T, shell 40 mm, heads 38 mm	SA-516 Gr 70 + 3 mm 316L cladding, polished inside A	—	Design 15.2 bar g / 93 °C and full vacuum (evacuated to 0.03 bar a). PSV-100 15 bar g. Operating at 2.0 h: 8.62 bar g, 57 °C; highest 8.88 bar g (cold batch, Transition tab). Shell governed by external pressure (jacket + vacuum).
R-101 J	R-101 jacket	Conventional jacket with spiral baffle	Tempered-water loop (P-101A/B, CW and LP-steam injection)	5.2 D	93 A	7.10 C	40.9 M	9.3 C	118.5 m² wetted at charge level, annulus 75 mm	SA-516 Gr 70 A	—	Design 5.2 bar g from P&ID-001 note 3 (Safety layers TSV-110). Loop pressure at the P-101 discharge 7.1 bar g, shut-off 8.1 bar g (estimated from the 90 kW pump power): see the checks.
M-101	R-101 agitator	Top-entry agitator, double mechanical seal	Keeps the VCM droplets suspended	15.2 M	93 M	8.62	57.0	—	338.1 kW absorbed at the end of the batch	316L wetted parts	450 A (290 M)	Absorbed power (model, K.agitKW 290 kW basis): 290.2 kW at 2.0 h, 323.7 kW at the heat-release peak, 338.1 kW at the end point. Motor rating assumed.
E-101	Reflux Condenser	Vertical shell-and-tube, VCM condensing in the tubes	Condenses VCM boiled off R-101; condensate runs back	15.2 / FV (tubes) · 6.0 (shell) M	93 (tubes) · 70 (shell) M	8.62 M	56.1 M	—	UA 130 kW/K, A ≈ 216.7 m²	316L tubes and tubesheets, CS shell A	—	UA from the model (K.UAc). Duty at the reference-batch peak 1708.5 kW with 91.1 m³/h CW; FV-130 wide open 540 m³/h. Tube side open to R-101 without isolation: same design, protected by PSV-100.
VP-101	Vacuum Pump	Liquid-ring vacuum pump (cooled ring liquid)	Evacuates R-101 before the charge (XV-109)	3.5 A	65	-0.98 M	30.0	—	3240 m³/h suction, ultimate 0.03 bar a	316 SS wetted A	90 A	Pump-down of 141.5 m³ to 0.05 bar a in 10.2 min (cold batch run); $V/S \cdot \ln(P_0/P)$ = 7.9 min. Motor assumed.
D-103	Emergency Vent Tank	Horizontal drum	Receives PSV-100 and XV-108 discharges; quench water	7.0 A	93 A	0.00	30.0	150.0 A	≈ 150 m³	CS + 3 mm CA A	—	Size and design assumed; capacity for one full two-phase relief to be confirmed by the relief study (HAZOP R08, Relief & vent tab).
D-106	VCM Gasholder 1.25 bar a	Vertical gasholder	Receives degassing, inert vent and stripped VCM at 1.25 bar a	1.0 A	65	0.24 M	35.0	141.4 D	D 6 m × H 5 m	CS A	—	Common to R-101…R-104 (assumed). Operating pressure from the model (K.Prec).
C-101	VCM Recovery Compressor	Oil-free reciprocating compressor	Recovered VCM from D-106 to condensation and TK-100	12.0 A	120	7.00 A	90.0	—	Suction 1.25 bar a	CS impact-tested A	250 A	Common to the four reactors (assumed). Only the suction pressure is in the model (K.Prec); the rest assumed.
P-101A/B	Jacket Circulation Pumps	Centrifugal, end suction (1 running + 1 standby)	Tempered-water loop through the R-101 jacket	10.0 A	93	7.10 C	38.2 M	—	470.7 m³/h (130 kg/s) × 52.3 m	CS casing, 13Cr impeller A	110 A (90 M)	Flow from the model (K.Fc, FI-110 ≈ 470 m³/h). Head implied by the 90 kW of econ() with η 0.78 × 0.95: 5.1 bar. Auto-start of P-101B recommended (HAZOP R01).
D-102	Blowdown / Slurry Tank	Vertical agitated tank	Receives the degassed slurry of a batch (XV-106)	3.5 / FV A	93 A	0.30 M	55.0	230.9 D	D 7 m × H 6 m	304L SS A	45 A	One batch at the end point: 133 t, 122.8 m³ (53.2 % of the tank). Operating pressure from the model (K.Pbd). Vent to VCM recovery recommended (HAZOP R23).
P-102	Slurry Pump	Centrifugal slurry pump	D-102 to the stripping column T-201	10.0 A	93	5.00 A	55.0	—	≈ 90 m³/h	Duplex SS A	22 A	Not in the model: all values assumed (four batches of ≈ 125 m³ per 6.9 h cycle average ≈ 72 m³/h).
T-201	Slurry Stripping Column	Shared finishing unit	Common to R-101…R-104 (block on PFD-100)	—	—	—	—	—	D 2 m × H 14 m		—	Outside this document: listed for completeness, specified by the finishing-section documents.

Tag	Name	Type	Service	Design P bar g	Design T °C	Oper. P bar g	Oper. T °C	Volume m³	Size	Material	Motor kW (absorbed)	Notes
S-201	Dewatering Centrifuge	Shared finishing unit	Common to R-101…R-104 (block on PFD-100)	—	—	—	—	—			—	Outside this document: listed for completeness, specified by the finishing-section documents.
D-201	PVC Dryer	Shared finishing unit	Common to R-101…R-104 (block on PFD-100)	—	—	—	—	—			—	Outside this document: listed for completeness, specified by the finishing-section documents.

4 R-101 wall thickness (ASME VIII Div. 1, indicative)

Cylinder (UG-27): t = PR/(SE − 0.6P); 2:1 ellipsoidal head (UG-32): t = PD/(2SE − 0.2P); external pressure (UG-28) by Factor A from the Windenburg–Trilling relation that the chart is drawn from, B = A·E/2 capped at 105 MPa, Pa = 4B/(3 Do/t); heads by UG-33. Inputs: S 138 MPa (SA-516 Gr 70 at 93 °C, typical), E 1, CA 3 mm, slurry 1050 kg/m³ for the static head, unsupported length L = 7 m jacketed straight side + one third of the head depth = 7.38 m.

Design pressure at the bottom tangent	15.91 bar g (top 15.2 + static 0.71)
Shell, internal pressure	t = 26.1 + CA 3 = 29.1 mm → 30 mm plate
Shell, external pressure	6.21 bar needs 39.8 mm → 40 mm (a 30 mm shell holds only 2.88 bar)
Shell nominal	40 mm, governed by external pressure (jacket + vacuum); Pa 6.32 bar, A 6.01e-4
Head, internal / external	26.2 mm + CA + 10 % → 35 mm / 38 mm → 38 mm nominal
MAWP corroded (top)	18.5 bar g
Jacket plate	12 mm for 5.2 bar g
Weights (estimate)	empty 85 t, operating 228 t, hydrotest 237 t

5 E-101 rating

The model's relation (step(), E-101 block): $Q = C_{cw}(1 - e^{-UA/C})(T_{cond} - T_{cw})$, $UA = UA_c(1 - b)(1 - f)$, with UA_c 130 kW/K, b the share of the tubes blanketed by inerts and f the fouling. Rating point: reference-batch peak, T_{cond} 56.1 °C, CW 32 °C, b 9.8 %.

Duty vs CW flow, clean (model peak: 1709 kW at 91 m³/h)

CW m³/h	NTU	ε	Duty kW	CW out °C
25	4.04	0.982	687	55.7
50	2.02	0.867	1214	52.9
91	1.11	0.670	1708	48.2
150	0.67	0.490	2057	43.8
200	0.51	0.397	2220	41.6
300	0.34	0.286	2401	38.9
400	0.25	0.223	2499	37.4
540	0.19	0.171	2579	36.1

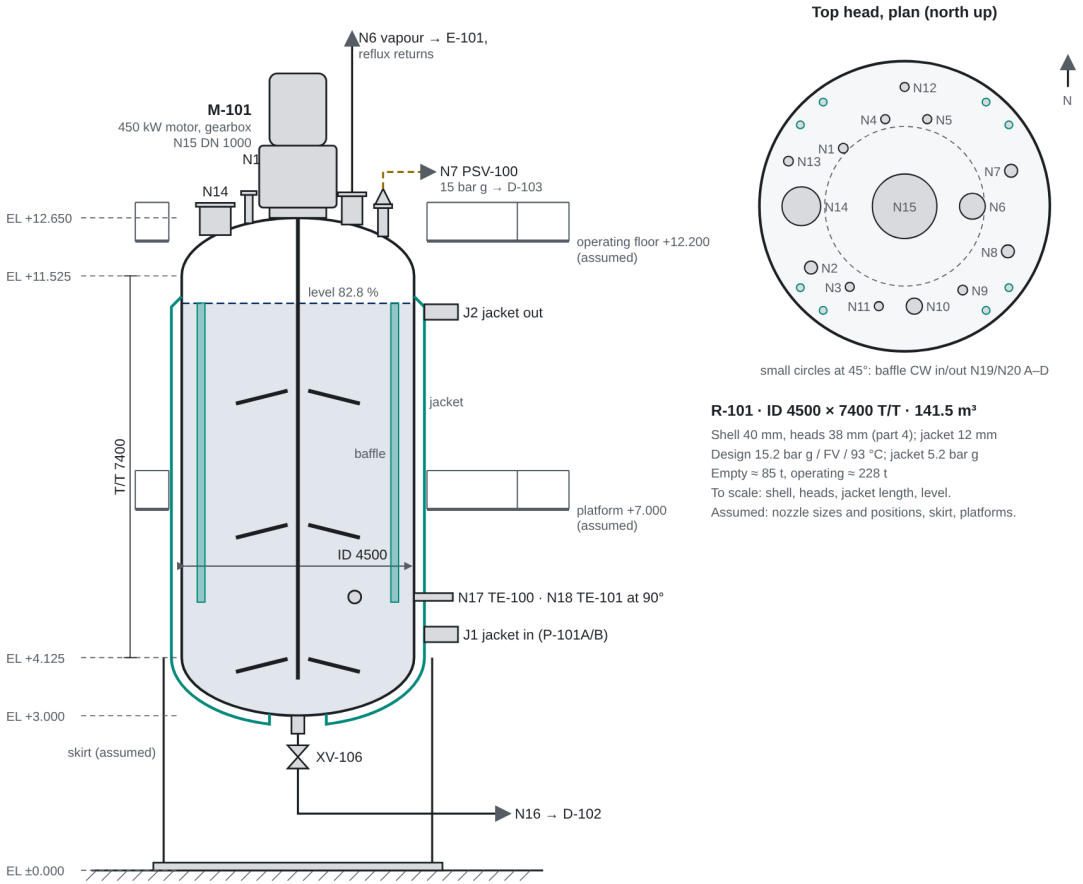
Duty vs fouling

Fouling f %	UA kW/K	Duty at FV-130 open kW	CW for the peak duty m³/h
0	117.3	2579	91
10	105.6	2342	106
20	93.8	2101	137
30	82.1	1855	234
45	64.5	1478	not reachable
60	46.9	1090	not reachable

6 R-101 nozzle list and general arrangement

Nozzle	Service	DN	Class	Location	Tag	Flow m³/h	v m/s	Basis
N1	VCM charge	150	150	Top head	FV-102 / XV-102	200	3.4	FIC-102 180 t/h
N2	Water + PVA charge	200	150	Top head	FV-101 / XV-101	240	2.3	FIC-101 240 m³/h
N3	Injection water (shrinkage)	50	150	Top head	FV-103	10	1.5	FIC-103 at FV-103 open
N4	Initiator	50	150	Top head	XV-103 (from P-104)	6	0.9	60-s dose (assumed volume)
N5	Short-stop	80	150	Top head	XV-104 (from D-105)	—	—	
N6	Vapour to E-101 / reflux return	400	150	Top head	—	1070	2.6	peak boil-up 22.8 t/h
N7	PSV-100 inlet (to D-103)	200	150	Top head	PSV-100	—	—	size by the relief study
N8	Emergency vent (to D-103)	200	150	Top head	XV-108	—	—	
N9	Degassing to VCM recovery	150	150	Top head	XV-105 / PV-105	—	—	
N10	Vacuum (to VP-101)	250	150	Top head	XV-109	3240	19.9	VP-101 suction volume (K.Svac)
N11	N_2 pad / blanket	50	150	Top head	PCV-141 / HV-141	—	—	1.5 bar g
N12	Pressure PT-100 / PT-101	50	150	Top head	PT-100, PT-101	—	—	two nozzles
N13	Level radar LT-100	150	150	Top head	LT-100	—	—	
N14	Manway	600	150	Top head	—	—	—	with davit
N15	Agitator mounting flange	1000	150	Top head, centre	M-101	—	—	double mechanical seal
N16	Bottom outlet (flush-bottom valve)	200	150	Bottom head, centre	XV-106	252	2.4	discharge at 1.5 bar Δp
N17	Thermowell TE-100 (control)	50	150	Shell, EL +5.3 m	TE-100	—	—	
N18	Thermowell TE-101 (SIS)	50	150	Shell, EL +5.3 m (90° from N17)	TE-101	—	—	
N19A–D	Baffle CW inlets (4)	80	150	Top head	FV-120	50	3.0	FIC-120 80 m³/h / 4 baffles at FV-120 open
N20A–D	Baffle CW outlets (4)	80	150	Top head	TI-121	50	3.0	
J1	Jacket inlet	300	150	Jacket, bottom	TI-110	471	2.0	P-101 470.7 m³/h
J2	Jacket outlet	300	150	Jacket, top	TI-111	471	2.0	
J3/J4	Jacket vent / drain	50	150	Jacket top / bottom	—	—	—	

Nozzle sizes and ASME B16.5 class 150 (17.7 bar g at 93 °C for group 1.1, typical) are assumed; flows from the model where it has them.



R-101 general arrangement (sketch, not to scale for nozzles; elevations assumed).

7 Checks against the model and the other documents

33 pass, 3 fail, 1 not available when this document was generated.

Check	Value	Criterion	Result	Source
R-101 design P ≥ PSV-100 set	15.2 vs 15 bar g	design ≥ set (UG-134: set ≤ MAWP)	✓	model header, K.Ppsv
R-101 design P ≥ highest operating P	15.2 vs 8.88 bar g	design ≥ max operating (cold batch, Transition tab)	✓	DS_REF / TR_D
R-101 design P ≥ PSHH-101 trip + margin	15.2 vs 12.5 bar g	design ≥ trip + 10 %	✓	K.Phh

Check	Value	Criterion	Result	Source
R-101 design P ≥ rule on the hottest grade	15.2 vs 13.6 bar g	max(1.1 P, P + 1.7) on 11.9 bar g at 70 °C	✓	Design tab rule, DS_IN Tmax
R-101 design T ≥ highest batch T + margin	93 vs 58.2 + 25 °C	design ≥ max + 25 °C (typical)	✓	DS_REF / TR_D
R-101 design T ≥ relieving temperature	93 vs 80.3 °C	contents reach the PSV set at this T (VCM + water + 0.25 bar inerts)	✓	PsV, PsW, K.Ppsv
R-101 volume from geometry = model VTOT	141.55 vs 141.55 m³	|Δ| < 0.1 %	✓	emGeom vs model
Fill level at charge = model level	82.8 vs 82.5 %	|Δ| < 1 % of height (recipe volumes at 57 °C)	✓	K.Wch, K.Mch; DS_REF.st
Fill level at charge below LSHH-100	82.8 vs 94 %	level < trip	✓	K.Lhh
E-101 UA on the datasheet = model	130 kW/K	= K.UAc	✓	K.UAc
E-101 rating reproduces the model at the peak	1708 vs 1709 kW	|Δ| < 2 %	✓	DS_REF.pk
E-101 capacity at FV-130 open ≥ peak duty	2579 vs 1709 kW	capacity ≥ duty	✓	K.Fcw130
M-101 motor ≥ absorbed power (peak and end)	450 vs 338 kW	rated ≥ 1.1 × max absorbed	✓	DS_REF pk/end Qag
M-101 motor ≥ JIAH-100 alarm	450 vs 420 kW	the high alarm sits inside the motor rating	✓	ALM (page)
Jacket design P ≥ loop pressure (P-101 running)	5.2 vs 7.1 bar g	CW return 2 bar g + P-101 Δp 5.1 bar	✗	econ() 90 kW, assumed η
Jacket design P ≥ P-101 shut-off	5.2 vs 8.1 bar g	blocked-in loop with the pump running	✗	assumed shut-off ratio
Jacket design P ≥ CW supply pressure	5.2 vs 4 bar g	CW injection cannot over-pressure the jacket	✓	assumed header
Jacket design T ≥ highest jacket inlet T	93 vs 86.6 °C	steam heat-up of the cold batch	✓	TR_D cold batch
R-101 shell holds jacket + vacuum (UG-28, indicative)	Pa 6.32 vs 6.21 bar	shell 40 mm, L 7.38 m	✓	emAsme
R-101 MAWP (corroded) ≥ design P	18.5 vs 15.2 bar g	MAWP at the top ≥ design	✓	emAsme
P-103 shut-off ≤ R-101 design P	17.6 vs 15.2 bar g	storage at 50 °C, shut-off = suction + 1.2 Δp	✗	K.Pvcm, PsV; assumed curve
D-102 holds one batch	122.8 vs 231 m³	batch volume ≤ 80 % of the tank	✓	DS_REF.end, PFD-100
TK-100 holds one VCM charge	56.5 vs 77.8 m³	charge ≤ 85 % of the drum	✓	K.Mch, PFD-100
E-101 tube side design = R-101 design	15.2 bar g / 93 °C	open to R-101 without isolation, protected by PSV-100	✓	P&ID-001
PSV-100 set ≤ design P of R-101	15 vs 15.2 bar g	set ≤ MAWP (design P used)	✓	ER_RLB (Relief & vent)
RD-100 set vs protected item	15 bar g	protects "PSV-100 inlet (polymer plugging)"	—	ER_RLB (Relief & vent)
TSV-110 set ≤ design P of R-101 J	5.2 vs 5.2 bar g	set ≤ MAWP (design P used)	✓	ER_RLB (Relief & vent)
P-103 motor rating = load list	75 vs 75 kW	same rating in both documents	✓	EE_LL (Electrical)
TK-107 motor rating = load list	2.2 vs 2.2 kW	same rating in both documents	✓	EE_LL (Electrical)
P-104 motor rating = load list	1.5 vs 1.5 kW	same rating in both documents	✓	EE_LL (Electrical)
M-101 motor rating = load list	450 vs 450 kW	same rating in both documents	✓	EE_LL (Electrical)
VP-101 motor rating = load list	90 vs 90 kW	same rating in both documents	✓	EE_LL (Electrical)
C-101 motor rating = load list	250 vs 250 kW	same rating in both documents	✓	EE_LL (Electrical)
P-101A/B motor rating = load list	110 vs 110 kW	same rating in both documents	✓	EE_LL (Electrical)
D-102 motor rating = load list	45 vs 45 kW	same rating in both documents	✓	EE_LL (Electrical)
P-102 motor rating = load list	22 vs 22 kW	same rating in both documents	✓	EE_LL (Electrical)
R-101 operating P = HMB stream 10	8.62 vs 8.62 bar g	|Δ| < 0.1 bar	✓	EP_HMB (Process engineering)

8 Engineering queries raised

Subject	Finding	Raised against	Proposed
R-101 jacket design pressure	P&ID-001 note 3 gives the jacket 5.2 bar g, but the loop runs at ≈ 7.1 bar g at the P-101 discharge (CW return 2 bar g + 5.1 bar implied by the 90 kW pump power in the model) and ≈ 8.1 bar g at shut-off.	PVC-PID-001 note 3; PVC-V1-ME-EQL-001; model econ() P-101 power	Raise the jacket design to 10 bar g (typical for tempered-water jackets) and re-check the shell for external pressure (it then needs a thicker shell or stiffening), or confirm a P-101 head below 3.2 bar and lower the pump power in the model.
R-101 shell thickness governed by external pressure	A 30 mm shell (internal pressure, UG-27) buckles under jacket 5.2 bar g + full vacuum (Pa 2.9 bar < 6.2 bar). The shell needs 40 mm with a 7.4 m unstiffened jacket.	PVC-V1-ME-EQL-001 R-101 datasheet	Vendor to choose between a thicker shell, jacket closure/stiffening rings or a half-pipe jacket; the wall resistance enters the jacket U (model U0 0.75 kW/m² K): confirm U with the final wall.
P-103 shut-off pressure above the R-101 design pressure	With TK-100 at 50 °C the VCM charge pump can reach ≈ 17.6 bar g at shut-off, above the 15.2 bar g design of R-101: a liquid-full reactor on the VCM charge is an overpressure case.	PVC-V1-PE-RLB-001 (relief basis); PVC-V1-ME-EQL-001	Add the liquid-overfill case (P-103 at shut-off) to the relief basis of PSV-100, or limit the pump curve / add a P-103 discharge PSV; LSHH-100 stays the first protection.
Equipment not in the model	Sizes, design conditions and motors of TK-107, D-104, P-104, D-105, D-103, D-106, C-101, P-102 and the motor ratings of M-101, P-101A/B, VP-101 are assumed in this document.	PVC-V1-ME-EQL-001	Confirm with the process data sheets (Process engineering tab) and the relief basis (D-103 volume, HAZOP R08).

Appendix A · Datasheets

A.1 R-101 PVC suspension reactor (vessel, jacket, agitator)

General

Service	Suspension polymerisation of VCM, batch; 1 of 4 identical reactors (R-101…R-104)
Type	Vertical, jacketed, stirred; 2:1 ellipsoidal heads; 4 water-cooled baffles; reflux condenser E-101 on the top head
Code	ASME VIII Div. 1 (typical)
Inside diameter × T/T	4500 × 7400 mm (model)
Total volume	141.5 m³ (geometry) = 141.5 m³ model VTOT
Working volume / level at charge	121.2 m³ → 82.8 % of the height (model start of reaction 82.5 %, LIC-100 SP 83 %)

Design conditions

Internal design pressure	15.2 bar g at the top + static head (15.91 bar g at the bottom tangent) (model header)
External design pressure	Full vacuum (evacuated to 0.03 bar a) + jacket 5.2 bar g → 6.21 bar
Design temperature	93 °C; MDMT −10 °C (MDMT assumed)
Relief	PSV-100 set 15 bar g to D-103 (see the Relief & vent tab)
Materials	Shell, heads: SA-516 Gr 70 + 3 mm 316L cladding, polished (Ra ≤ 0.4 µm) to limit polymer build-up; jacket SA-516 Gr 70 (assumed)
Corrosion allowance	3 mm on the base metal (jacket-water side); cladding not credited (assumed)
Joint efficiency / radiography	E = 1 (full RT) (assumed)
Shell thickness	40 mm nominal: 30 mm for internal pressure, 40 mm for external pressure → governed by external pressure (jacket + vacuum)
Head thickness	38 mm nominal (incl. 10 % forming): 35 mm internal, 38 mm external (bottom head jacketed)
MAWP (corroded, at the top)	18.5 bar g
Hydrotest	≈ 19.8 bar g at the top (UG-99(b): 1.3 × MAWP × stress ratio 1) (typical)

Operating conditions

Design state (2.0 h into the reaction)	57.0 °C, 8.62 bar g, X 26 %, Qrx 2.71 MW, level 82.9 % (snapDesign)
Reference batch	end point at 5.27 h, X 84.8 %, peak 3.43 MW at X 72 %, highest 8.69 bar g / 57.33 °C
Charge per batch	60 t water (55 °C) + PVA, 51 t VCM (25 °C), initiator; 133.0 t slurry at the end point

Jacket

Type	Conventional jacket with spiral baffle over the bottom head and 7 m of the straight side (type assumed; length from the model)
Heat-transfer area	118.5 m² wetted at the charge level (model jacketArea); U clean 0.75 kW/m² K (model)
Medium / circulation	Tempered water, P-101A/B 130 kg/s (471 m³/h); CW injection TV-110B, LP steam injection TV-110A

Operating (2.0 h)	inlet 38.2 °C, outlet 40.9 °C, duty 1499 kW
Design	5.2 bar g (P&ID-001 note 3) / 93 °C (T assumed); jacket plate 12 mm

Baffles

Cooled baffles	4 × water-cooled, 22 m² total, U 0.8 kW/m² K, CW FIC-120 80 m³/h (model); design as the vessel, external 15.2 bar g

Agitator M-101

Type	Top-entry, multi-stage impellers, double mechanical seal with pressurised barrier fluid (Safety layers SEAL) (assumed)
Absorbed power	290 kW at 2.0 h, 324 kW at the peak, 338 kW at the end point (model, K.agitKW 290 kW basis); 2.40 kW/m³
Motor	450 kW, fixed speed, Ex d/e for Zone 1 (assumed); JIAH/JIAL-100 alarms

Weights (estimate)

Empty / operating / hydrotest	85 t / 228 t / 237 t (steel from the thicknesses above + 25 % + 12 t agitator)

A.2 E-101 reflux condenser

General

Service	Condenses the VCM boiled off R-101; condensate drains back by gravity; inerts vented at the top (HIC-135)
Type	Vertical fixed-tubesheet shell-and-tube (TEMA BEM), VCM condensing inside the tubes, CW on the shell side (assumed)

Performance

UA clean	130 kW/K (model K.UAc)
Overall U / area	0.6 kW/m² K (Design tab, assumed) → 217 m²
Tubes	25.4 mm OD × 4 m, ≈ 679 tubes, triangular pitch 1.25 d; bundle ≈ 909 mm (bundle-diameter correlation) (assumed)
Design state (2.0 h)	duty 971 kW, CW 36 m³/h, CW out 55.0 °C, inert blanketing 9.7 % of UA
Reference-batch peak	duty 1709 kW, CW 91 m³/h 32 → 48.2 °C, condensing 56.1 °C, blanketing 9.8 %
Maximum CW	FV-130 wide open 540 m³/h; anti-foam limit 12 m³/h per % of XI-100 conversion (model)

Design

Tube side (VCM)	15.2 bar g / FV / 93 °C, same as R-101 (no isolation between them)
Shell side (CW)	6.0 bar g / 70 °C (assumed; CW header 4.0 bar g)
Materials	Tubes and tubesheets 316L; shell CS; CA 0 / 3 mm (assumed)
Alarms	TIAH-131 CW return 58 °C (scaling), TIAL-132 top cold 46 °C (inerts) (ALM)

A.3 P-101A/B jacket circulation pumps

Service	Tempered-water loop R-101 jacket; one running, one standby (auto-start recommended, HAZOP R01)
Type	Centrifugal, end suction, back pull-out, mechanical seal (assumed)
Rated flow	130 kg/s = 471 m³/h at 40 °C (model K.Fc; FI-110 ≈ 470 m³/h)
Differential	5.1 bar = 52 m, from 90 kW electrical (model econ()) × η motor 0.95 × η pump 0.78 (η assumed)
Suction / discharge	2 bar g (CW return, assumed) / 7.1 bar g; shut-off ≈ 8.1 bar g
Fluid temperature	38.2 °C pump discharge at 2.0 h; 86.6 °C at the end of steam heat-up (cold batch)
Motor	110 kW rated (assumed); absorbed 90 kW (model)
Trips and alarms	FIAL-110 300 m³/h, FSLL-110 < 120 m³/h while reacting (30 s) → trip A
Design	Casing 10 bar g / 93 °C (assumed)

A.4 D-102 blowdown / slurry tank

Service	Receives the degassed slurry of one batch through XV-106; feeds P-102 to the stripping column T-201
Size	D 7 m × H 6 m (PFD-100) = 231 m³
Batch received	133.0 t, 122.8 m³ at the end point (reference batch); discharge permitted below 2.0 bar g in R-101
Operating	0.3 bar g (model K.Pbd), ≈ 55 °C
Design	3.5 bar g / FV / 93 °C (assumed)
Agitator	45 kW, keeps the slurry suspended (assumed)
Material	304L SS (assumed)
Vent	To VCM recovery, enclosed (HAZOP R23, Safety layers)

A.5 D-103 emergency vent (dump) tank

Service	Receives PSV-100 and the emergency vent XV-108 of R-101; quench water to stop a relieved runaway (recommended)
Volume	≈ 150 m³ (assumed; to be confirmed by the relief study, HAZOP R08)
Design	7 bar g / 93 °C (assumed)
Operating	Atmospheric, vent to treatment
Material	CS, 3 mm CA (assumed)
Notes	Two-phase relief from a runaway (DIERS) sets the volume; see the Relief & vent tab and the HAZOP report.